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RENTMATE

Your Smart Home Away From Home

Student Accommodation · Roommate Matching · PG Discovery

PRODUCT REQUIREMENTS DOCUMENT

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PLATFORM	Web Application (MERN Stack)
PROGRAM	Gryork TechPreneur Training Program
TEAM	FantasticFour
STATUS	Final

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1. Executive Summary

RentMate is a student-centric accommodation discovery platform that helps students find verified PGs, hostels, shared apartments, and compatible roommates. The platform simplifies accommodation discovery through intelligent search, college-based recommendations, roommate compatibility analysis, interactive maps, and expense management.

RentMate aims to build a trusted ecosystem where students can safely relocate to a new city without depending on brokers, social media groups, or unreliable referrals.

2. Problem Statement

Students relocating to different cities for education face serious difficulties finding affordable, safe, and trustworthy accommodation. Current methods rely heavily on Facebook Groups, WhatsApp Communities, local brokers, and word-of-mouth referrals.

Issue	Description
Fake Listings	Fraudulent postings with misleading information
Hidden Charges	Undisclosed fees revealed only after commitment
Poor Roommate Matching	No structured way to find compatible housemates
Lack of Verification	No identity or property verification mechanisms
Unsafe Environments	No trust signals for evaluating safety
Time-Consuming Searches	Manual browsing across fragmented sources

3. Vision Statement

To become the most trusted student accommodation ecosystem by simplifying housing discovery, roommate matching, and student relocation experiences.

4. Goals & Objectives

4.1 Product Goals

Goal	Objective
Simplify Accommodation Discovery	Help students find housing quickly
Improve Roommate Matching	Connect compatible students
Increase Transparency	Provide verified listings
Reduce Search Time	Minimise effort and time spent
Enhance Student Safety	Build trust through verification
Improve User Experience	Create a seamless platform

4.2 Business Objectives

- Build a scalable student housing marketplace
- Increase verified property inventory
- Improve user trust and engagement
- Enable sustainable platform growth

5. User Personas

The platform is designed around three primary user types. Each role has distinct needs, pain points, and responsibilities.

Persona	Needs	Pain Points
Student	Affordable accommodation near college, safe environment, compatible roommates, verified listings	Fake listings, hidden costs, unknown roommates, time-consuming searches
Property Owner	Better visibility, more inquiries, higher occupancy	Vacant rooms, limited reach to student market
Admin	Fraud prevention, listing moderation, user management, platform analytics	Scalable verification, content quality control

6. User Stories

ID	User Story
US-1	As a student, I want to search accommodation near my college so I can find housing quickly.
US-2	As a student, I want verified listings so I can avoid scams.
US-3	As a student, I want roommate recommendations so I can find compatible housemates.
US-4	As a student, I want to save properties for later comparison.
US-5	As a student, I want to manage shared expenses with roommates.
US-6	As a property owner, I want to publish listings so students can discover my property.
US-7	As an admin, I want to verify listings to prevent fraudulent content.

7. System Overview

RentMate serves three major stakeholders with distinct capabilities:

Capability	Student	Property Owner	Admin
Register / Login	Yes	Yes	Yes
Search Accommodation	Yes	—	—
View Property Details	Yes	—	—
Save Favourite Properties	Yes	—	—
Find & Match Roommates	Yes	—	—
Manage Shared Expenses	Yes	—	—
Contact Property Owner	Yes	—	—

Create / Edit / Delete Listings	—	Yes	—
View Student Inquiries	—	Yes	—
Verify Listings	—	—	Yes
Manage Users	—	—	Yes
Remove Fraudulent Content	—	—	Yes
View Analytics Dashboard	—	Yes	Yes

8. Project Scope

The scope of RentMate defines the features and functionalities included in the MVP while clearly identifying features intentionally excluded from the current release.

8.1 In Scope

Module	Included Features
User Management	Student & owner registration/login, profile management, email & phone verification
Accommodation Discovery	Search by city, locality, college name; property detail pages; availability status
Advanced Filtering	Budget, type, occupancy, amenities, distance from college
Property Listing Mgmt	Create, edit, delete listings; image upload; vacancy management
Roommate Matching	Profile creation, lifestyle preferences, compatibility score, discovery
Verification System	Student & property verification, verified badge, admin approval workflow
Wishlist & Favorites	Save properties & roommate profiles; view saved items
Expense Splitter	Add/categorise shared expenses, calculate shares, view outstanding balances
Reviews & Ratings	Property ratings, review submission & display (verified students only)
Notifications	Vacancy alerts, new listings, price drops, roommate match notifications
Administration Panel	User management, listing verification, analytics dashboard, content moderation

8.2 Explicit Non-Goals

To keep the MVP focused and deliverable, the following are intentionally excluded from scope:

Category	Excluded Items
Online Payments	Rent gateway, UPI integration, security deposit collection
Legal Documentation	Digital rental agreements, e-signatures, lease generation
Advanced AI	AI chatbot, fraud detection, recommendation engine
Communication	Voice/video calling, group messaging
Mobile Applications	Native Android & iOS apps
Social Features	Community forums, student feed, event management

9. Core Features

A. Smart Accommodation Search

Students can search properties using city, area, college name, budget, property type, and sharing type.

B. Verified Property Listings

Each listing includes owner details, rent, deposit, distance from college, amenities (WiFi, AC, laundry, mess, CCTV, gym, parking, power backup), and property/room images.

C. Smart Roommate Matching

Students build roommate profiles covering sleep schedule, study hours, food preference, smoking preference, cleanliness level, introvert/extrovert tendency, and noise tolerance. The system generates a compatibility score with explanatory reasons.

Example Output
Compatibility Score: 92%
Reasons: Similar sleeping schedule • Similar study habits • Same food preference

D. Interactive Maps

Maps display PG location, college, nearby hospitals, grocery stores, bus stops, cafes, and ATMs via Google Maps API.

E. Expense Splitter

Supports categories: Rent, Electricity, Water, Internet, Groceries. Calculates per-person shares and shows outstanding balances between roommates.

F. Verification System

Verification Type	Method
Student Verification	College ID, Email, Phone
Property Verification	Owner badge, Admin approval workflow

G. Notification System

Triggers: New property added, price drop, vacancy available, roommate request accepted.

H. Review & Rating System

Students rate properties on food quality, safety, internet connectivity, and cleanliness. Only verified students may submit reviews.

10. Functional Requirements

10.1 Student Features

ID	Requirement
FR-1	User Registration
FR-2	Login Authentication
FR-3	Search Accommodation

FR-4	Budget Filter
FR-5	Location Filter
FR-6	College-Based Search
FR-7	Property Details View
FR-8	Save Favourite Properties
FR-9	Contact Property Owner
FR-10	Roommate Discovery
FR-11	Compatibility Score
FR-12	Expense Splitter
FR-13	Profile Management

10.2 Property Owner Features

ID	Requirement
FR-14	Create Listing
FR-15	Edit Listing
FR-16	Delete Listing
FR-17	Manage Availability
FR-18	View Student Inquiries

10.3 Admin Features

ID	Requirement
FR-19	Verify Listings
FR-20	Manage Users
FR-21	Remove Fraudulent Content
FR-22	Analytics Dashboard

11. Unique Selling Points (USP)

#	USP	Description
1	Student-First Design	Built exclusively for the student accommodation use case
2	Smart Roommate Matching	Compatibility scoring based on lifestyle preferences
3	College-Based Search	Find PGs directly linked to a specific college
4	Verified Ecosystem	Trust signals via verified owner & student badges
5	Expense Management	Built-in splitter for shared monthly expenses

6	Interactive Maps	Live neighbourhood context around every property
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12. System Modules

Module	Pages / Components
Public Pages	Home, About, Contact, Property Listings
Student Module	Register, Login, Dashboard, Profile, Saved Properties, Expenses, Notifications
Property Owner Module	Add Property, Manage Properties, Analytics
Admin Module	Dashboard, User Management, Listing Verification

13. User Flow

The student journey through the platform follows a clear sequential flow:

Step	Action	Description
1	Register	Create student account and verify identity
2	Complete Profile	Add college, preferences, and roommate criteria
3	Select College	Anchor search to a specific institution
4	Search PG	Browse verified listings in the area
5	Apply Filters	Narrow by budget, type, occupancy, amenities
6	View Property	Review full details, images, map, and reviews
7	Save Property	Add to wishlist for later comparison
8	Find Roommate	Discover compatible students seeking accommodation
9	Check Compatibility	Review compatibility score and lifestyle alignment
10	Contact Owner	Send inquiry directly to the property owner
11	Move In	Complete the offline agreement and relocate

14. Non-Functional Requirements

Category	Requirement
Performance	Search results returned in under 2 seconds
Availability	99% uptime SLA
Scalability	Support 1,000+ concurrent users
Security	JWT authentication and secure REST APIs
Reliability	Accurate and consistent data across sessions
Compatibility	Mobile, tablet, and desktop browser support

Maintainability	Modular and scalable MERN architecture
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15. Success Metrics

Category	Metric	Target
User	Registered Students	Growth Month-over-Month
User	Daily Active Users (DAU)	Tracked
User	Monthly Active Users (MAU)	Tracked
User	User Retention Rate	Tracked
Marketplace	Number of Verified Listings	Growing inventory
Marketplace	Inquiry-to-Conversion Rate	Tracked
Product	Search Response Time	< 2 seconds
Product	Profile Completion Rate	> 80%
Product	Roommate Match Success Rate	Tracked
Satisfaction	Average User Rating	> 4 / 5
Satisfaction	Positive Feedback Rate	> 80%

16. Market Analysis

Audience Type	Segment
Primary	Undergraduate Students, Postgraduate Students, International Students, Internship Students
Secondary	PG Owners, Hostel Owners, Property Managers

The student accommodation market continues to grow due to increasing student migration across cities for higher education and internships. Existing solutions lack student-focused roommate matching and verified accommodation discovery, creating a clear opportunity for RentMate.

17. Competitive Analysis

Feature	RentMate	Generic Housing Apps	Social Media Groups
Student Focused	Yes	No	No
Verified Listings	Yes	Partial	No
Roommate Matching	Yes	No	No
Compatibility Score	Yes	No	No
College-Based Search	Yes	No	No
Expense Splitter	Yes	No	No
Interactive Maps	Yes	Partial	No

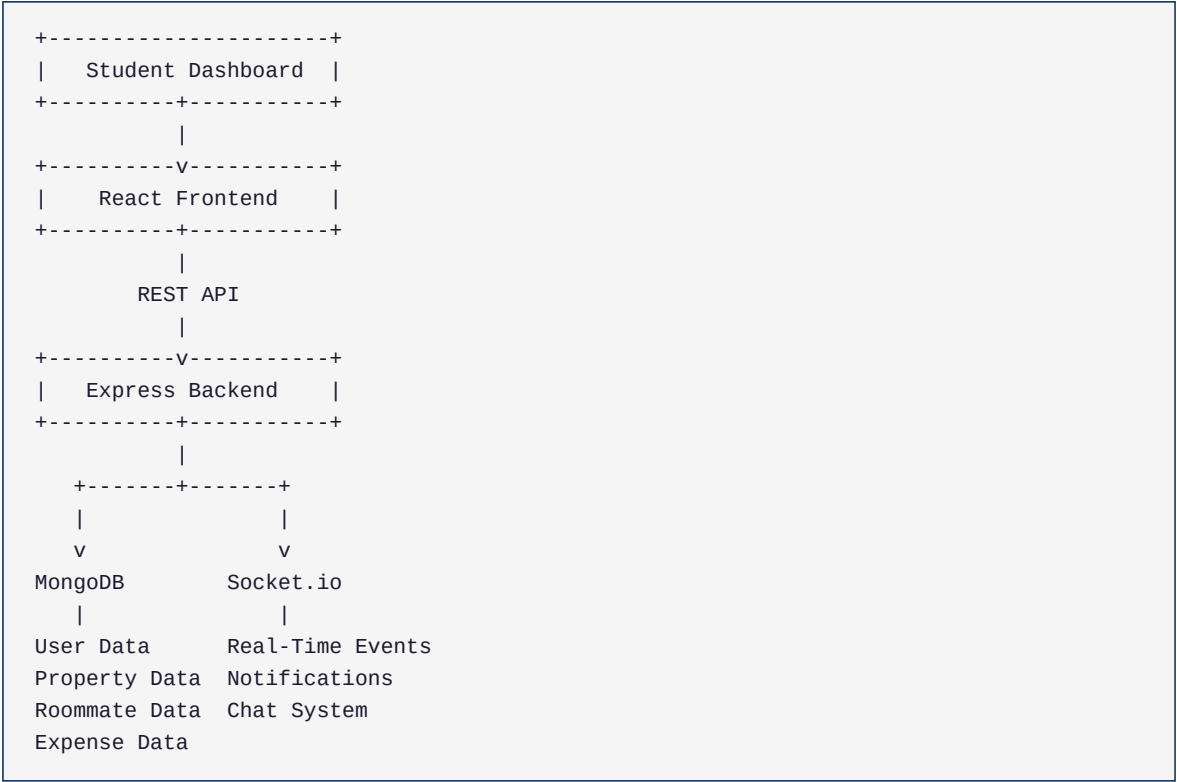
Admin Moderation	Yes	Partial	No
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18. Technical Architecture

18.1 Technology Stack

Layer	Technology	Responsibility
Frontend	React.js + Tailwind CSS + Framer Motion	User, owner, and admin interfaces; animated, responsive UI
Backend API	Node.js + Express.js	Authentication, business logic, REST endpoints
Database	MongoDB	Users, properties, roommates, expenses, reviews
Authentication	JWT	Stateless, role-based session tokens
Real-Time	Socket.io	Live notifications, chat, and event broadcasting
Maps	Google Maps API	Interactive property and neighbourhood maps
Media	Cloudinary	Property and profile image hosting

18.2 High-Level Architecture Diagram



19. Database Design

Users Collection

Field	Type
UserID	ObjectId

Name	String
Email	String
Role	String
College	String
Phone	String

Properties Collection

Field	Type
PropertyID	ObjectId
OwnerID	ObjectId
Rent	Number
Deposit	Number
Location	String
Amenities	Array

Roommates Collection

Field	Type
UserID	ObjectId
Preferences	Object
CompatibilityScore	Number

Expenses Collection

Field	Type
ExpenseID	ObjectId
Category	String
Amount	Number
CreatedAt	Date

Reviews Collection

Field	Type
ReviewID	ObjectId
PropertyID	ObjectId
Rating	Number
Comment	String

20. UI/UX Design Guidelines

Principle	Detail
Design Style	Modern startup style with glassmorphism elements and smooth animations
Interface	Minimalistic, student-friendly, accessibility-focused
Responsiveness	Mobile-first design; supports mobile, tablet, and desktop
Theming	Dark mode, light mode, and glassmorphism elements
Core Pages	Home, Search, Property Details, Dashboard, Expense Tracker, Profile

21. Assumptions & Constraints

Category	Item
Assumption	Students have reliable internet access
Assumption	Property owners provide accurate listing information
Assumption	Users maintain updated profiles over time
Assumption	Students are willing to share roommate lifestyle preferences
Constraint	Limited project timeline (MVP scope)
Constraint	Web application only — no native mobile apps in MVP
Constraint	No payment gateway integration in MVP
Constraint	Limited manual moderation resources

22. Future Roadmap

Phase	Features
Phase 2	Mobile Application (Android & iOS), Community Forums, Enhanced Chat System, Improved Reviews & Ratings
Phase 3	AI Roommate Matching, AI Property Recommendations, Fraud Detection Models, Rent Payment Gateway, Advanced Analytics

23. Expected Outcome

RentMate will provide a trusted, transparent, and student-centric accommodation ecosystem that simplifies housing discovery, improves roommate matching, reduces fraud, and enhances the relocation experience for students.

By combining accommodation search, verification systems, compatibility-based roommate matching, and expense management into a single platform, RentMate can evolve from a project prototype into a scalable startup solution.

— API Endpoints

Group	Method	Endpoint
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Authentication	POST	/api/auth/register
Authentication	POST	/api/auth/login
Authentication	POST	/api/auth/logout
Properties	GET	/api/properties
Properties	GET	/api/properties/:id
Properties	POST	/api/properties
Properties	PUT	/api/properties/:id
Properties	DELETE	/api/properties/:id
Roommates	GET	/api/roommates
Roommates	POST	/api/roommates/preferences
Roommates	GET	/api/roommates/matches
Expenses	POST	/api/expenses
Expenses	GET	/api/expenses
Expenses	PUT	/api/expenses/:id
Expenses	DELETE	/api/expenses/:id
Reviews	POST	/api/reviews
Reviews	GET	/api/reviews/:propertyId
Notifications	GET	/api/notifications
Notifications	PUT	/api/notifications/read